

PRACTICAL 11

Aim : List different adapters in any computer/ laptop and configure network connection in your institute/ premises LAN.

Input :

Different Network Adapters

A computer/laptop may have the following network adapters:

- **Ethernet adapter (LAN) :** Wired connection using an RJ-45 cable.
- **Wi-Fi adapter (Wireless LAN) :** Connects to wireless networks.
- **Bluetooth adapter :** Provides short-range wireless communication.
- **USB network adapter :** External Ethernet or Wi-Fi adapter connected through USB.
- **Virtual network adapter :** Created by software such as VPNs or virtual machines.
- **Fiber-optic network adapter/ NIC :** Used for high-speed fiber-based network connections.

Configuring a LAN Connection in an Institute/Premises

1. Identify the network adapter

Open Command Prompt and run : `ipconfig /all`

2. Connect the computer to the LAN

- For a wired LAN, connect an Ethernet cable from the computer to the network switch/wall socket.
- For Wi-Fi, select the institute's authorized wireless network and enter the provided credentials.

3. Configure IPv4

Open:

Control Panel → Network and Internet → Network and Sharing Center → Change adapter settings

Then:

Right-click Ethernet/Wi-Fi → Properties → Internet Protocol Version 4 (TCP/IPv4) → Properties

Use the settings provided by the institute's network administrator.

For a DHCP network, select:

- Obtain an IP address automatically
- Obtain DNS server address automatically

For a static-IP network, enter the administrator-provided:

- IP address
- Subnet mask
- Default gateway
- Preferred/alternate DNS servers

4. Test the connection

Run:

`ipconfig`

`ping 127.0.0.1`

`ping <default-gateway>`

`ping 8.8.8.8`

`ping google.com`

5. Check the routing table

`route prints`